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Bioassay and biochemical analyses of insecticide resistance in southern African *Anopheles funestus* (Diptera: Culicidae)

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Abstract

Anopheles funestus Giles has been implicated as a major malaria vector in sub-Saharan Africa where pyrethroid insecticides are widely used in agriculture and public health. Samples of this species from northern Kwazulu/Natal in South Africa and the Beluluane region of southern Mozambique showed evidence of resistance to pyrethroid insecticides. Insecticide exposure, synergist and biochemical assays conducted on *A. funestus* suggested that elevated levels of mixed function oxidases were responsible for the detoxification of pyrethroids in resistant mosquitoes in these areas. The data suggested that this mechanism was also conferring cross-resistance to the carbamate insecticide propoxur.

Introduction

The World Health Organization (WHO) estimates approximately 500 million malaria cases per annum resulting in two to three million deaths (WHO, 1996). More recent estimates for the African region suggest approximately one million deaths resulting from 250 million cases. Mortality in African children ranges from 0.43–0.68 million of the ~66 million children aged 0–4 years infected each year (Snow *et al.*, 1999). The major malaria vector mosquitoes in sub-Saharan Africa are *Anopheles gambiae* Giles and *A. arabiensis* Patton of the *A. gambiae* species complex and *A. funestus* Giles (Diptera: Culicidae) (Gillies & De Meillon, 1968).

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Malaria vector control directed at indoor resting mosquitoes began in the 1930s with the use of pyrethrum extracted from flowers (Park Ross, 1936). DDT and dieldrin with their residual effects replaced pyrethrum as an indoor vector control strategy shortly after the second world war. However, reports of resistance to these insecticides in target mosquito populations soon appeared. Davidson (1956) reported on the development of dieldrin resistance in a population of *A. gambiae* from northern Nigeria. This mode of resistance showed cross-resistance to related compounds such as BHC and chlordane but did not confer resistance to DDT. Further reports of dieldrin and DDT resistance in both *A. gambiae* and *A. arabiensis* (see Coetzee *et al.*, 1999) coupled with environmental concerns led to the total banning of dieldrin and discontinued use of DDT in most regions. Pyrethroid insecticides and, to a lesser extent, carbamate and organophosphate insecticides, have subsequently found a range of applications in African countries affected by

malaria. These applications include agricultural and public health use. Pyrethroid-based indoor residual spraying (IRS) and insecticide-treated nets and curtains (ITNs) are currently advocated as standard malaria vector control strategies (review by Phillips, 2001). The emergence of resistance to pyrethroids in a number of target African mosquito populations has highlighted the importance of research into the effect of resistance on malaria transmission including the underlying mechanisms causing the resistance (Hemingway & Ranson, 2000). Identification of mechanisms already implicated in insecticide resistance may give an indication of the potential for the evolution of resistance to alternative insecticides.

Pyrethroid / DDT insecticide resistance has been described in *A. gambiae* in parts of west Africa (Elissa *et al.*, 1993; Chandre *et al.*, 1999). Reduced target site sensitivity arising from a single point mutation in the sodium channel gene has been implicated as the predominant mode of resistance in these populations (Martinez-Torres *et al.*, 1998) although increased levels of detoxifying enzymes may also be playing a role (Brooke *et al.*, 1999). Target site insensitivity (Ranson *et al.*, 2000) and elevated levels of detoxifying mixed function oxidases (MFO) (Vulule *et al.*, 1999) have been implicated in an increase in permethrin tolerance in *A. gambiae* in Kenya. Techniques designed to determine the role of mixed function oxidases in pyrethroid resistance include the use of the synergist piperonyl butoxide (pbo) which serves as an inhibitor of mixed function oxidase activity (Brogdon & McAllister, 1998). Biochemical techniques have been designed to quantify levels of mixed function oxidases and other detoxifying enzyme levels and activities (Penilla *et al.*, 1998).

Anopheles funestus was the major vector of malaria in South Africa prior to the early 1950s (Gillies & De Meillon, 1968; Gillies & Coetzee, 1987). This species was eradicated from the region when widespread malaria control using DDT as an indoor residual insecticide was implemented (see Coetzee & Hunt, 1998). Since then it has been reported only once from an unsprayed area where it was quickly eliminated again by indoor spraying with DDT (De Meillon *et al.*, 1977). Health authorities in northern Kwazulu / Natal in South Africa discontinued the use of DDT for malaria vector control in 1996 although it was retained for use in at least one other province. The pyrethroid deltamethrin was used as a replacement indoor residual insecticide. Since then the number of reported malaria cases in northern Kwazulu / Natal has risen dramatically (unpublished minutes of the South African Ministry of Health Malaria Advisory Group, 2000). Recently, male and female members of *A. funestus* were collected from exit window traps attached to houses that had previously been sprayed with deltamethrin (Hargreaves *et al.*, 2000). Progeny from a number of egg batches from wild-caught females showed varying levels of pyrethroid resistance when exposed to WHO discriminating dosages of permethrin (Hargreaves *et al.*, 2000).

Anopheles funestus is one of nine members of the *A. funestus* group and one of two species within this group implicated in malaria transmission, the second being *A. rivulorum* Leeson which is implicated as a minor vector in Tanzania (Wilkes *et al.*, 1996). The *A. funestus* infection rate can exceed that of other vector species (Temu *et al.*, 1998) and has been recorded at 5.4% for *Plasmodium falciparum* (Welch) (Plasmodidae) in northern Kwazulu / Natal (Hargreaves *et al.*, 2000). *Anopheles funestus* is highly anthropophilic and endophilic making it susceptible to indoor spraying of residual insecticide. Hargreaves *et al.* (2000) showed that four other members of the group, *A. rivulorum*, *A. leesoni* Evans, *A. vaneedeni* Gillies and Coetzee and *A. parensis* Gillies can also be found resting indoors. These are all described as predominantly exophilic and zoophilic although both *A. vaneedeni* and *A. rivulorum* will feed on humans in the absence of suitable animal hosts. All *A. funestus* group members breed in well-vegetated, permanent pools, swamps and slow-moving streams (Gillies & de Meillon, 1968).

This paper presents the results of bioassay, synergist and biochemical analyses of samples of *A. funestus* from South Africa and southern Mozambique.

Materials and methods

Field collections

Mosquitoes collected in Ndumu, northern Kwazulu / Natal province, South Africa, were taken from exit window traps of sprayed houses early each morning (see Hargreaves *et al.* (2000) for locality data). Collections from the Beluluane region of southern Mozambique (25°93'S; 32°52'E) in the vicinity of an aluminium smelter plant were done by hand searches using an aspirator and torch each morning in unsprayed houses during February to April, 2000.

Wild mosquitoes were assayed using WHO discriminating dosages of insecticides. The remaining live and all dead specimens were transported to the insectaries of the South African Institute for Medical Research in Johannesburg. Females were induced to lay eggs and subsequent families were reared through to adults. F1 progeny from each family were divided into subsamples according to the number emerged. One subsample was stored in liquid nitrogen to be used for biochemical analysis while the other subsamples were used for insecticide and synergist bioassays.

Species identification

Anopheles funestus group wild specimens and those from families reared from wild females were identified to species level using the polymerase chain reaction–single strand conformation polymorphism (PCR–SSCP) procedure of Koekemoer *et al.* (1999) which discriminates between *A. funestus*, *A. vaneedeni*, *A. rivulorum* and *A. leesoni*.

Insecticide bioassays

All insecticide bioassays described in this report were carried out according to standard WHO specifications using WHO insecticide susceptibility kits and insecticide treated papers (WHO, 1998). All batches of insecticide treated papers used to test wild-caught mosquitoes or F1 progeny (see table 1 for details) were tested on a laboratory colony (SKUQUA) of *A. quadriannulatus* Theobald (Diptera: Culicidae) known to be susceptible to all insecticides used in this study. Any batches of papers showing less than 100% mortality in these control tests were discarded.

Between five and 30 mosquitoes were assayed per paper in each trial. Control assays included exposures of samples

Table 1. WHO bioassay results on wild-caught members of the *Anopheles funestus* group and *A. gambiae* complex from Beluluane, Mozambique.

Insecticide	*A. funestus* group 24 h post-exposure		*A. gambiae* complex 24 h post-exposure	
	% mortality	n	% mortality	n
0.05% deltamethrin	48	29	100	4
0.05% lambdacyhalothrin	51	88	100	33
4% DDT	100	53	untested	–
0.1% propoxur	56	97	100	20
0.1% bendiocarb	95	82	100	13
1% fenitrothion	100	97	100	13
Untreated control papers	0	51	0	20

PCR identification (Scott *et al.*, 1993) of a subsample of the *A. gambiae* complex mosquitoes revealed the presence of both *A. arabiensis* and *A. merus* Dönitz. Controls: susceptible *A. quadriannulatus* SKUQUA colony.

of mosquitoes to papers treated only with the solvent of the insecticide in question or to untreated papers. All exposures lasted 1 h except for the fenitrothion assays which lasted 2 h. Knockdown was recorded every 10 min and final mortality was recorded after a 24 h holding period during which time a 10% sucrose solution was made available to survivors.

Cross-resistance bioassays

Beluluane F1 familial progeny survivors following exposure to deltamethrin or propoxur were pooled and allowed to rest for a period of 72 h. Each sample was then divided into two equal subsamples. One subsample was re-exposed to deltamethrin while the other was re-exposed to propoxur using WHO treated papers. Mortality between the two subsamples was compared after a 24 h holding period.

Synergist studies

The synergist piperonyl butoxide, an inhibitor of mixed function oxidases, was used in conjunction with the pyrethroid lambdacyhalothrin and the carbamate propoxur in a subset of bioassays conducted on wild-caught Beluluane *A. funestus*, a subset of F1 progeny from two Beluluane wild caught *A. funestus* females and a mixed sample of F1 progeny from eight Beluluane wild females. Each subset was divided into two equal samples. One sample was pre-exposed to 4% piperonyl butoxide impregnated filter paper for 1 h and then both samples were immediately exposed to 0.05% lambdacyhalothrin or 0.1% propoxur for 1 h. Final mortality after a 24 h holding period was compared between synergized and unsynergized samples.

Biochemical analysis

Biochemical assays were used to quantify levels of mixed function oxidases, non-specific esterase and glutathione-S-transferase (GST) activity, as well as to detect the presence of an altered acetylcholinesterase in individual mosquitoes from as many *A. funestus* families as possible. Batches of 47 one- to three-day-old mosquitoes were assayed per microtitre plate according to the method described by Penilla *et al.* (1998). Each plate contained a sample of F1 progeny from familial material (previously stored in liquid nitrogen) as well as a sample of at least ten SKUQUA mosquitoes for comparative purposes.

Data from each microtitre plate were standardized by equalizing average enzyme levels / activity of the SKUQUA sample on each plate to a predetermined value and adjusting the F1 familial data accordingly. This was necessary because temperature fluctuations in the laboratory gave different rates of enzyme reaction leading to disparate absorption readings between plates. Hence, values of enzyme level obtained did not reflect absolute amounts of enzyme per mosquito but average levels per family relative to base-line SKUQUA levels. Statistical analysis involved two sample t-tests by means. The adjusted mean enzyme level / activity of families from each locality were compared to the corresponding SKUQUA means.

Results

Collections from the Beluluane region of southern Mozambique yielded large numbers of mosquitoes of the *A. funestus* group and *A. gambiae* complex. Fifty to sixty specimens of members of the *A. funestus* group and 10–50 of the *A. gambiae* complex were collected per house. WHO insecticide susceptibility tests using a range of insecticides at standard concentrations were performed. The results of these tests are summarized in table 1. Only the *A. funestus* group showed evidence of resistance to pyrethroids and the carbamate propoxur.

Table 2 shows WHO deltamethrin and propoxur bioassay results performed on F1 progeny from families reared from wild-caught females from the Beluluane region. All families listed were identified by PCR–SSCP analysis as *A. funestus* s.s. Mortality ranged from 0% to 91% across the deltamethrin assays and from 0% to 86% across the propoxur assays. There was a significant correlation in percentage mortality between deltamethrin and propoxur assays by family (R^2 = 0.844, P = 0.008, d.f. = 6).

Data presented in table 3 detail the cross-resistance bioassay results of two samples of pooled F1 *A. funestus* that had survived either the deltamethrin or the propoxur exposures described in table 2. Chi-square analysis shows no significant difference in mortality in second exposure results between exposure to deltamethrin or to propoxur (P = 0.19).

The effect of the mixed function oxidase synergist piperonyl butoxide on lambdacyhalothrin and propoxur

Table 2. WHO bioassay results of subsamples of F1 progeny from families raised from female *Anopheles funestus* (identified by SSCP analysis) collected in Beluluane, Mozambique.

Family number	0.05% deltamethrin 24 h post-exposure		0.1% propoxur 24 h post-exposure	
	% mortality	n	% mortality	n
M98	9	11	18	11
M88	29	7	–	–
M82	91	11	86	7
M8	0	6	–	–
M76	11	9	–	–
M73	29	7	–	–
M70	25	4	–	–
M7	0	12	17	12
M64	33	12	14	7
M58	0	13	0	14
M48	57	14	18	11
M32	0	13	9	11
M28	9	11	0	5
M22	0	6	–	–
M15	82	11	–	–
M127	10	10	–	–
Control papers (solvent treated)	14	64	3	29

Controls: susceptible *A. quadriannulatus* SKUQUA colony.

Table 3. WHO bioassay results of cross-resistance experiments using *Anophleles funestus* F1 survivors from previous Beluluane, Mozambique family exposures.

	Second exposure to:	
Survivors of	0.05% deltamethrin 24 h % mortality (n)	0.1% propoxur 24 h % mortality (n)
0.05% deltamethrin	46 (22)	55 (20)
0.1% propoxur	19 (16)	18 (17)

Survivors were divided into two samples. Sample size denoted by figures in parentheses.

susceptibility in (from Beluluane) *A. funestus* is detailed in table 4. The difference in mortality (24 h post-exposure) between synergized and unsynergized subsamples was highly significant across all lambdacyhalothrin samples ($P < 0.0001$) and significant across propoxur samples ($P = 0.01$).

Biochemical analysis of subsamples of *A. funestus* F1 progeny from listed Ndumu families (see Hargreaves *et al.*, 2000) showed that the mean level of mixed function oxidase across these families was significantly higher than that of the corresponding SKUQUA mean ($P = 0.001$) based on a two sample t test for means. Sample sizes per family detailed in figs 1 and 2 ranged from 4 to 23. Figure 1 shows the relative level of mixed function oxidase per family with the SKUQUA base line set at zero. There was no significant correlation between mixed function oxidase level and permethrin bioassay mortality in these families ($R^2 = –0.347$, $P = 0.245$**).** The mean level of glutathione S-transferase activity across these families was not significantly higher than that of the corresponding SKUQUA mean ($P = 0.5866$) although some families did show high activity levels (fig. 2). There was no significant correlation between glutathione S-transferase activity level and permethrin bioassay mortality in these families ($R^2 = 0.05$, $P = 0.872$).

Biochemical analysis of subsamples of *A. funestus* F1 progeny from listed Beluluane families (see table 2) shows that the mean level of mixed function oxidase across these families was also significantly higher than that of the corresponding SKUQUA mean ($P = 0.003$). Sample sizes per family detailed in figs 3 and 4 ranged from 5 to 21. Figure 3 shows the relative level of mixed function oxidase per family with the SKUQUA base-line set at zero. There was no significant correlation between mixed function oxidase level and deltamethrin bioassay mortality in these families ($R^2 = 0.172$, $P = 0.508$). The mean level of glutathione S-transferase activity in these families (fig. 4) was not significantly higher than that of the corresponding SKUQUA mean ($P = 0.7918$) and there was no significant correlation between glutathione S-transferase activity and deltamethrin bioassay mortality ($R^2 = 0.25$, $P = 0.332$).

Mean non-specific esterase levels, using paranitrophenyl acetate as a substrate, across families from Ndumu and Beluluane were significantly lower than that of the corresponding SKUQUA mean ($P = 0.012$ and $P = 0.00001$ respectively). Acetylcholinesterase assays gave no suggestion of reduced propoxur inhibition in families from both localities. There is a significant negative correlation between mixed function oxidase level and glutathione S-transferase activity across families from both localities ($P = 0.0089$) although the correlation is weak ($R^2 = –0.163$).

Discussion

Subsamples of *A. funestus* F1 progeny originating from both collection localities showed similar biochemical profiles for the enzyme systems assayed compared to the base-line *A. quadriannulatus* (SKUQUA) samples. Most families gave indications of elevated mixed function oxidase levels while some families also showed relatively high glutathione S-transferase activity. The negative correlation between mixed function oxidase levels and glutathione S-transferase activity suggests that the physiological burden associated with increased enzyme activity precludes the possibility of one individual carrying higher than normal levels of both enzyme systems simultaneously. The implications in terms of resistance are unclear. Glutathione S-transferases may serve as a secondary or minor detoxifying agent in resistant *A. funestus* or their elevation may preclude high levels of mixed function oxidase mediated resistance. There are currently no published records implicating glutathione S-transferases in pyrethroid resistance. The mixed function

Table 4. WHO lambdacyhalothrin (pyrethroid) and propoxur (carbamate) bioassay results comparing piperonyl butoxide (pbo) synergized and unsynergized subsamples of *Anopheles funestus* from Beluluane, Mozambique.

Sample	24 h % mortality			
	4% pbo + 0.05% lambdacyhalothrin	0.05% lambdacyhalothrin	4% pbo + 0.1% propoxur	0.1% propoxur
A. funestus group (wild-caught)	97 (36)	55 (40)	–	–
A. funestus group (wild-caught)	98 (44)	48 (48)	–	–
A. funestus M70	100 (9)	25 (4)	–	–
A. funestus M48	100 (10)	57 (14)	–	–
A. funestus Mixed sample 1	–	–	100 (21)	79 (24)
A. funestus Mixed sample 2	–	–	94 (34)	81 (26)
A. funestus Control	0 (19)		0 (39)	

Control mosquitoes were exposed to pbo followed by exposure to papers treated with the relevant insecticide solvent only. Figures in parentheses denote sample size.

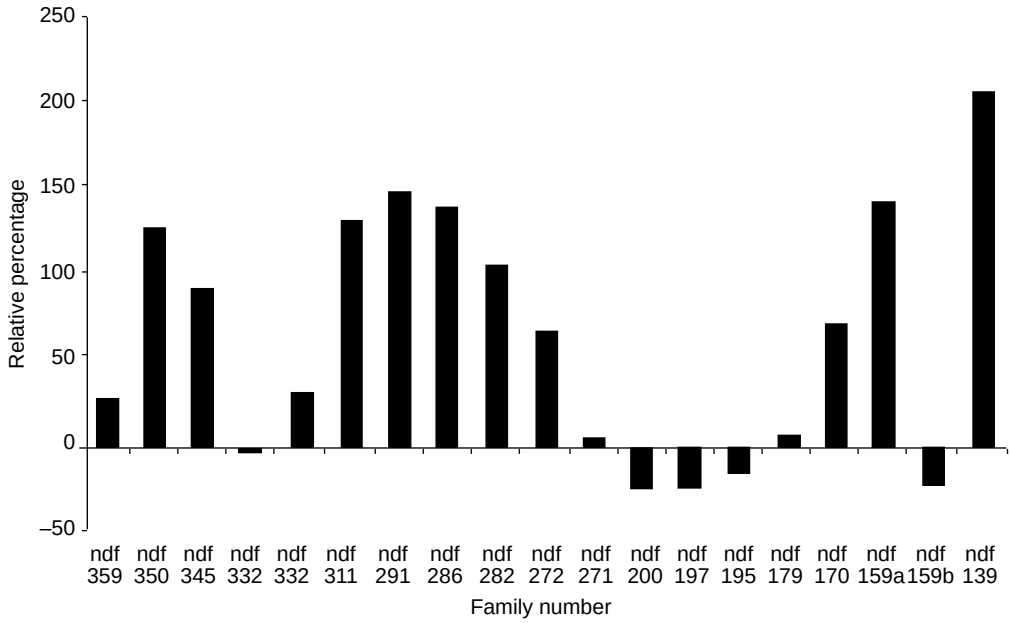

Fig. 1. Percentage increase or decrease in mixed function oxidase levels of listed *Anopheles funestus* families from the Ndumu area, South Africa relative to the base-line SKUQUA level set at zero.

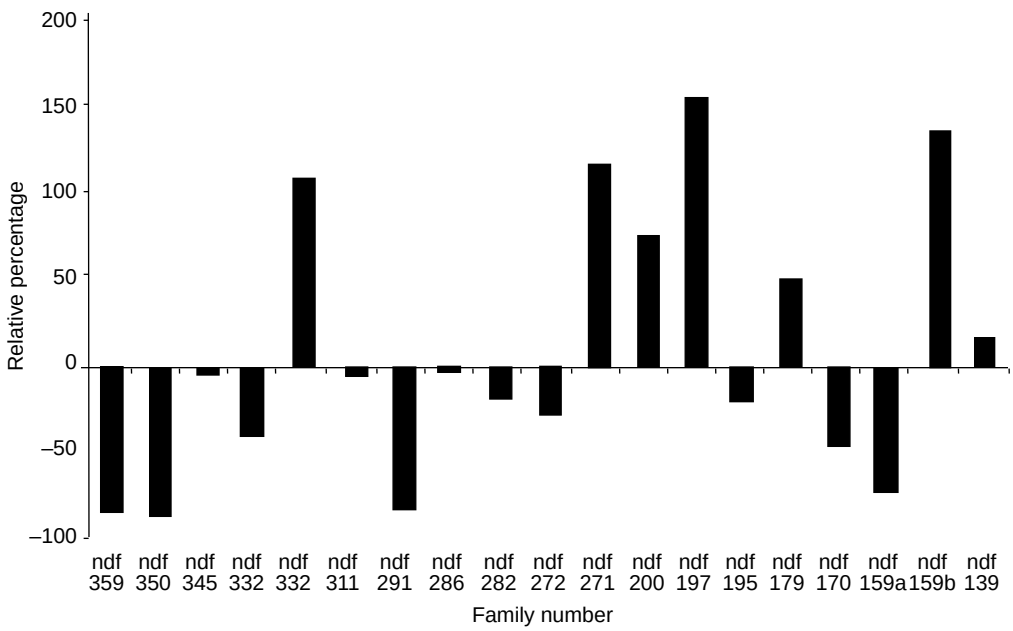

Fig. 2. Percentage increase or decrease in glutathione S-transferase activity of listed *Anopheles funestus* families from the Ndumu area, South Africa relative to the base-line SKUQUA level set at zero.

oxidase synergist piperonyl butoxide has not been implicated in glutathione S-transferase activity inhibition. It thus seems likely that glutathione S-transferases are not associated with pyrethroid/propoxur resistance in these samples and that their elevation as detected in some families serves some other unrelated physiological effect. The lack of significant correlation between mixed function oxidase level and pyrethroid bioassay mortality may be due to discrepancies in the amounts of mixed function oxidase detectable per sample depending on the length of time each

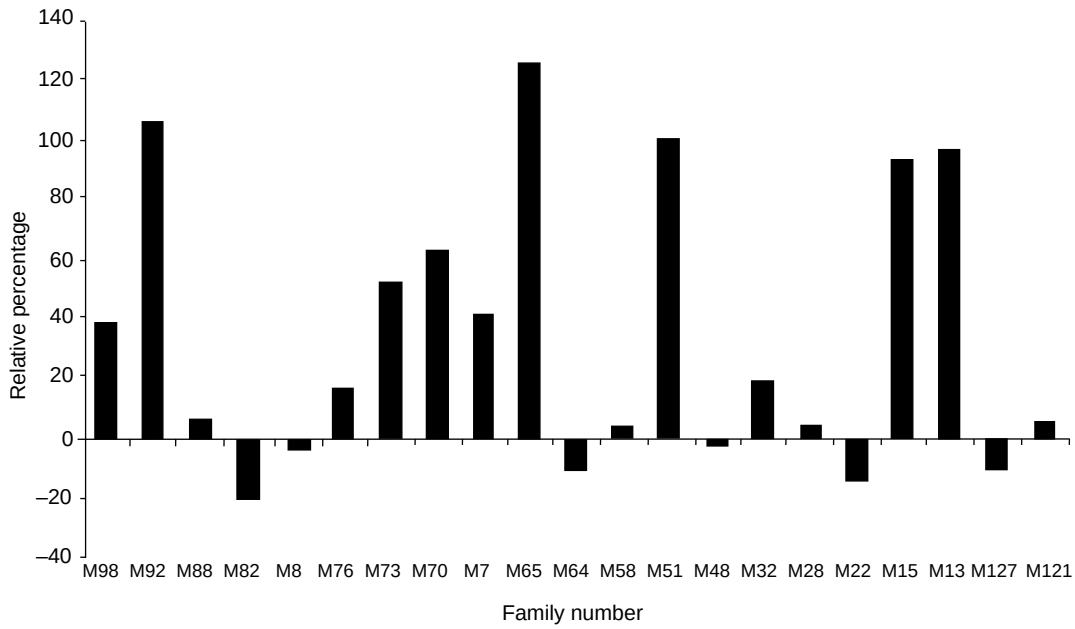

Fig. 3. Percentage increase or decrease in mixed function oxidase levels of listed *Anopheles funestus* families from the Beluluane area, southern Mozambique relative to the base-line SKUQUA level set at zero.

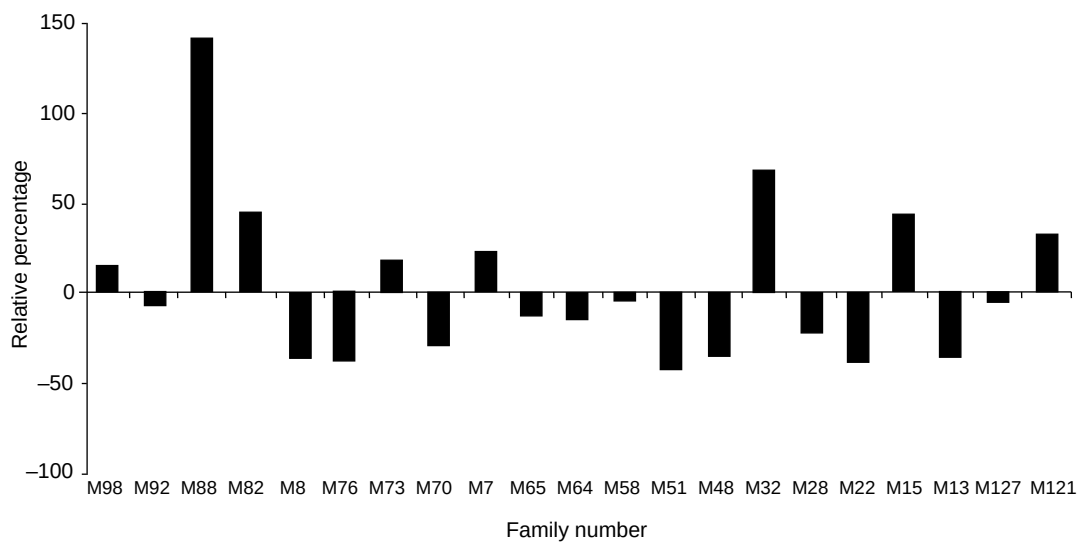

Fig. 4. Percentage increase or decrease in glutathione S-transferase activity of listed *Anopheles funestus* families from the Beluluane area, southern Mozambique relative to the base-line SKUQUA level set at zero.

sample was stored in liquid nitrogen, unequal distribution of resistance genotype frequency between subsamples of F1 progeny from each family and overall physiological condition of mosquitoes used in the respective assays.

Mixed function oxidases may be involved in the detoxification of almost all insecticides but are most notably associated with pyrethroid resistance (Hemingway & Ranson, 2000). The evidence for mixed function oxidase based cross-resistance between pyrethroids and the carbamate propoxur presented in this paper is compelling and unusual. Carbamate resistance has been closely associated with esterase metabolism (Hemingway *et al.*, 1998) while carbamate/organophosphate resistance has been associated with target site insensitivity following an alteration of the target site acetylcholinesterase (Hemingway & Georghiou, 1983). Cochran (1987) showed that piperonyl butoxide successfully synergized resistance to natural pyrethrins and the carbamate bendiocarb in several field strains of the German cockroach *Blattella germanica* (Linnaeus) (Dictyoptera: Blattellidae).

The presence of detectable numbers of *A. funestus* mosquitoes resting in houses sprayed with commercial formulations of deltamethrin in northern Kwazulu / Natal is considered to be a major causative factor in the recent upsurge of reported malaria cases in that region during the past few years. Other factors including anti-malarial drug resistance, heavy rainfall and cross border migration have also been identified (Phillips, 2001; unpublished minutes of the South African Ministry of Health Malaria Advisory Group, 2000). The implications for malaria vector control in Kwazulu / Natal and southern Mozambique are clear – an alternative vector control strategy needs to be identified. Health authorities in northern Kwazulu / Natal have recently re-sprayed those areas reporting high numbers of malaria cases with DDT as an interim measure (Kwazulu / Natal Malaria Control Programme unpublished report, 2000). Rotational or mixed use of insecticides (Curtis *et al.*, 1993), or a combination of indoor residual spraying and insecticide-treated bed nets are possible alternatives.

The impact of the recent re-use of DDT on malaria incidence in these areas has yet to be fully evaluated. Ongoing entomological surveys in the affected regions of northern Kwazulu / Natal have reported a single unfed female *A. funestus* specimen found dead inside an exit window trap. Two specimens identified as *A. rivulorum* have been reared from larval catches and one *A. funestus* female was caught feeding on a person indoors in an unsprayed house (K. Hargreaves, unpublished report). It therefore seems likely that the re-use of DDT in northern Kwazulu / Natal has dramatically reduced the *A. funestus* population, but not to the point where this species is undetectable.

Agricultural land use is often closely associated with an increase in the prevalence of vector-borne disease. Lacey & Lacey (1990) review associations between rice-growing activity and vector-borne disease prevalence. This review includes examples of the development of insecticide resistance in mosquito species of medical importance that breed in rice fields treated with insecticides. Roberts & Andre (1994) state that more than 90% of all insecticides produced have been used for agricultural purposes and that this has created serious problems in mosquito control programmes. Insecticide resistance in *A. funestus* is likely to have arisen from the indiscriminate and widespread use of pyrethroids in agriculture and not from public health usage of insecticides. *Anopheles funestus* larvae prefer well-vegetated permanent water bodies where insecticides might accumulate in small amounts leading to the development or enhancement of resistance in the larvae. Pyrethroid / DDT resistance in *A. gambiae* in Côte d'Ivoire is believed to have evolved from the early use of DDT and later use of pyrethroids in agriculture (Chandre *et al.*, 1999). Overlap of insecticide use in public health and the agricultural sector may lead to an increase in the prevalence of insecticide resistance in malaria vectors in the southern African region.

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